

WSI Viewer

Architecture & Developer Reference

Overview

The WSI Viewer is a browser-based whole-slide image viewer that streams SVS (and other OpenSlide-compatible TIFF) files directly from Google Cloud Storage via HTTP Range requests. It integrates with an external pathology API for slide metadata and supports Tumor Cell Annotation (TCA) overlays along with in-browser annotation tools.

Viewer URL pattern:

```
https://[BASE_URL]/wsi-viewer/?patient_id=[PATIENT_ID]&event_id=[EVENT_ID]&selected_slide_id=[SLIDE_ID]
```

1. Viewer URL and Request Flow

URL Parameters

All three query parameters are required to open the viewer:

Parameter	Description	Example
patient_id	Patient UUID	a1ce9f01-218d-4505-9906-957549121805
event_id	Event / encounter ID	189704
selected_slide_id	Slide to open first (block ID)	B1

Example URL:

```
https://viewer.example.com/wsi-viewer/?patient_id=a1ce9f01-218d-4505-9906-957549121805&event_id=189704&selected_slide_id=B1
```

Opening from Your Application

```
const url = 'https://' + baseUrl + '/wsi-viewer/?' + new URLSearchParams({
  patient_id: patientId,
  event_id:   eventId,
  selected_slide_id: slideId,
}).toString();
window.open(url, '_blank');
```

The application does not call the pathology API directly. The viewer backend handles all API communication using its own credentials (see Section 2).

End-to-End Request Flow

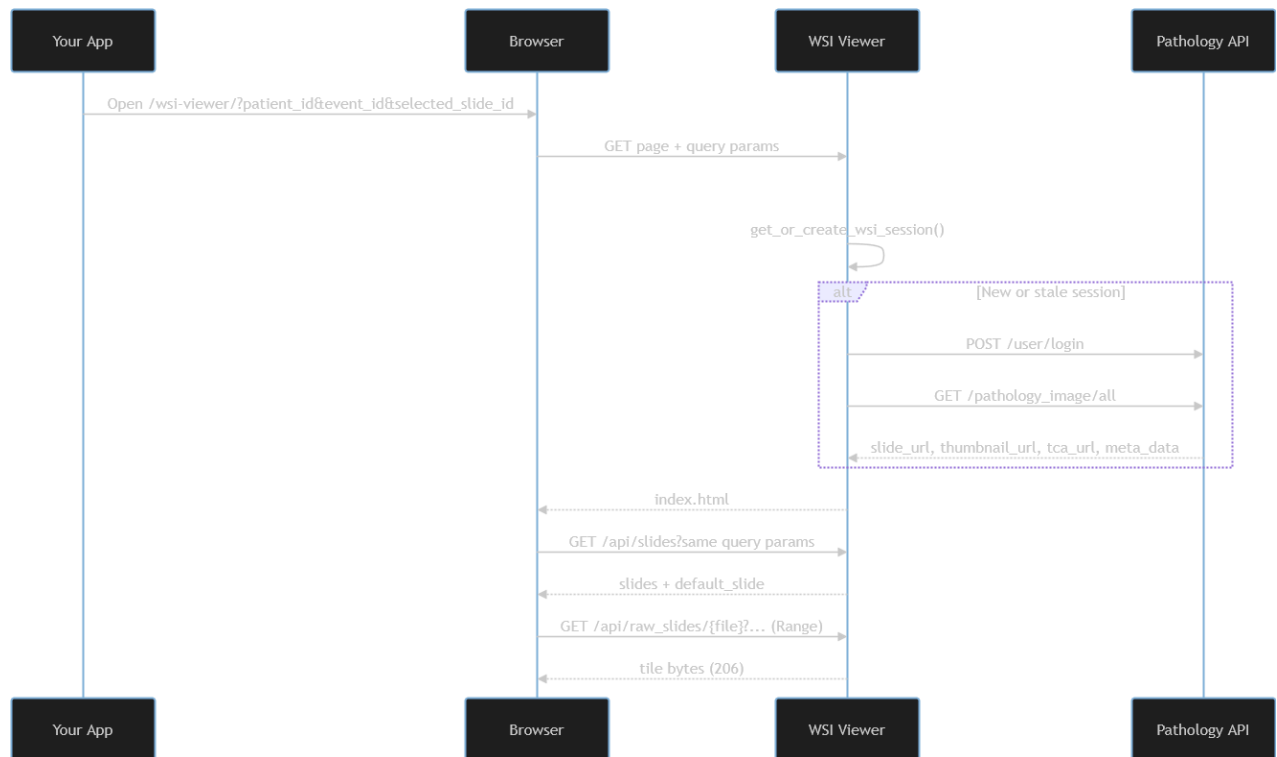


Figure 1 — End-to-end request flow from application open to tile streaming

Session Lifecycle

On first load, the server calls `get_or_create_wsi_session()` with an in-memory key:

```
| pid_<patient_id>_<event_id>_<selected_slide_id>
```

- Session is reused if younger than `WSI_URL_REFRESH_MINUTES` (default: 10 min).
- If stale: re-login + re-fetch `/pathology_image/all`, then rebuild with fresh signed URLs.
- Idle expiry: `SESSION_TTL` (default: 30 min).
- No auth token in the URL — session is identified by the three query parameters.

The frontend (`viewer.js`) detects WSI mode when `patient_id` is in the query string and sets `API_BASE` to the current path prefix. Every API call appends the same three query parameters:

```
| function apiPath(path) {
|   return API_BASE + path + '?' + API_QUERY; // patient_id, event_id, selected_slide_id
| }
```

API Routes

All routes require the three query parameters (`patient_id`, `event_id`, `selected_slide_id`):

Method	Path	Purpose
GET	/ (via <code>/wsi-viewer/</code>)	Viewer HTML
GET	<code>/api/slides</code>	Slide list + <code>default_slide</code>
GET	<code>/api/thumbnail/{slide_name}</code>	Thumbnail image

GET / HEAD	/api/raw_slides/{filename}	Slide streaming (Range)
GET	/api/slide-direct-url/{filename}	Optional signed URL for browser
GET	/api/overlay-config/{slide_name}	TCA overlay config
GET	/api/overlay-file/{filename}	TCA overlay assets

Deployment Paths

Layer	Typical path
Browser	/wsi-viewer/?patient_id=...
Gateway	Routes to viewer service
ROOT_PATH env	e.g. /in/wsi-viewer — stripped before handlers run

2. External Pathology API

The viewer calls the pathology/EMR API from the server only. Users never see the authToken or API credentials.

Configuration

Credentials are read from integrate_config.json, overridden by environment variables:

Variable	Purpose
EXTERNAL_API_BASE_URL	API base URL
EXTERNAL_API_EMAIL	Login email
EXTERNAL_API_PASSWORD	Login password

Example integrate_config.json:

```
{ "base_url": "https://api.example.com",
  "email": "service@example.com",
  "password": "secret" }
```

Missing config returns HTTP 503 when a new session is needed.

Session API Calls

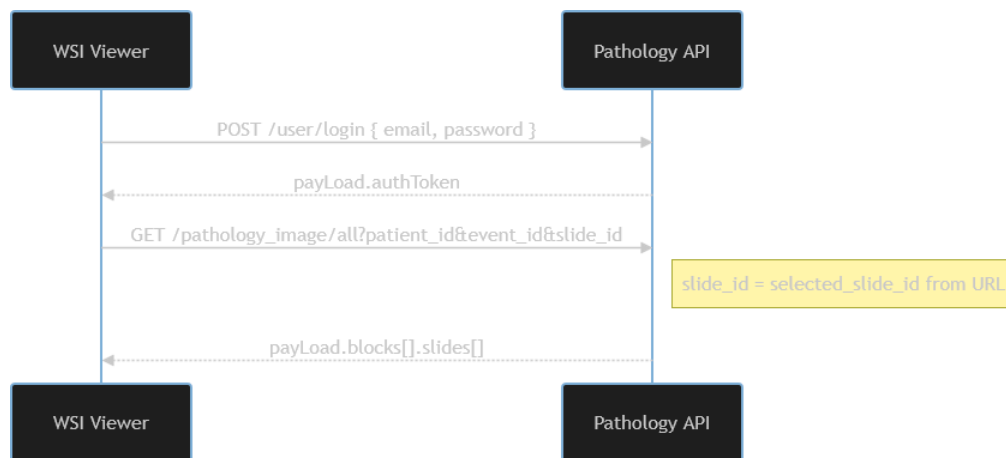


Figure 2 — Pathology API calls made per session

Expected API Response

The GET /pathology_image/all endpoint returns a blocks array with slides:

```
{ "payload": { "selected_slide_id": "B1",
  "blocks": [{ "block_id": "A1", "slides": [{
    "slide_id": "B1",
    "slide_url": "https://storage.googleapis.com/bucket/1087-25.svs",
    "thumbnail_url": "https://storage.googleapis.com/bucket/1087-25_thumbnail.png",
    "tca_url": "https://storage.googleapis.com/bucket/1087-25_tca.zip",
    "status": "FINISHED",
    "meta_data": { "width": "112000", "height": "86000",
      "mpp": "0.2527", "objective_power": "40", "vendor": "aperio" }
  }]}] } }
```

How the Viewer Builds the Session

- Collect every slide with a non-empty slide_url.
- Put the slide matching selected_slide_id first (becomes default_slide in the UI).
- Store slide_paths, api_slides (full objects), and thumbnail_paths.
- Signed slide_url values are refreshed when the session is older than WSI_URL_REFRESH_MINUTES to keep tile loading working.

3. Streaming Slides from GCS

The `slide_url` from the API is typically an HTTPS GCS URL (public or signed). The viewer does not download the full SVS file to disk. The browser uses `GeoTIFFTileSource` with HTTP Range requests; the backend proxies ranges when needed (CORS, signed URLs).

```
Browser --Range: bytes=a-b--> /api/raw_slides/1087-25.svs?patient_id=...
Viewer  --Range--> signed GCS URL
Browser <--206 Partial Content--
```

Raw Slide Endpoint

GET `/api/raw_slides/{filename}?patient_id=...&event_id=...&selected_slide_id=...`

- Proxies the Range header to the slide's `slide_url`.
- Returns 206 with Content-Range, Accept-Ranges: bytes, and CORS headers.
- HEAD provides Content-Length for tile source setup.

Optional: GET `/api/slide-direct-url/{filename}` returns the signed URL so the browser may read GCS directly. On failure the proxy URL is used.

```
const rawSlideUrl = apiPath('/api/raw_slides/' + slide.filename);
// GeoTIFFTileSource.getAllTileSources(url, { useRangeRequests: true })
```

Multi-page SVS (main + label + macro): the largest page is the main view; smaller pages appear as corner insets.

In DevTools: look for many small 206 responses with Range: bytes=... on `/api/raw_slides/...`

4. Overlays (TCA / Density)

Tumor Cell Annotation (TCA) is a server-provided heatmap per slide, toggled via the TCA button in the slide list. It is separate from user-drawn polygons (see Section 5).

Source

Each slide in the pathology API response may include `tca_url` — a URL to a zip file containing overlay assets. On first use the server downloads and extracts each zip into a per-slide temp folder.

File pattern	Role
*_density.png	Heatmap image
*_metadata.json	Grid / WSI dimensions for alignment
*_grid.json	Optional grid data

Slides without `tca_url` do not show the TCA button.

Overlay API Endpoints

GET `/api/overlay-config/{slide_name}` returns:

```
{ "available": true,
  "density_image": ".../api/overlay-file/{slide}_density.png?patient_id=...",
  "metadata": ".../api/overlay-file/{slide}_metadata.json?...",
  "slide_metadata": { "width", "height", "mpp", "objective_power", "vendor" } }
```

GET /api/overlay-file/{filename} serves the extracted PNG or JSON through the viewer (the browser never fetches tca_url directly). The available flag requires both density PNG and metadata JSON to be present.

UI Flow

- Slide list checks overlay config and shows or hides the TCA button per slide.
- On slide load: config is fetched, _overlayConfig and _slideMetadata are stored (used for measurements in micrometres).
- User clicks TCA: density image is overlaid on OpenSeadragon with adjustable opacity.

5. User Annotations

Polygons, text labels, and distance lines are browser-only — they are not saved to the server or the pathology API and are lost on page refresh.

Tool	Action
Draw polygon	Single-click to add vertices, double-click to finish (min 3 points), or exit draw mode to save
Add labels	Click point, enter text; label appears with anchor and offset
Measure	Two clicks to place a line; distance shown in micrometres if mpp is in slide metadata, otherwise pixels
Delete	Click a label, measure, or polygon, then confirm
Clear all	Removes all annotations on the current slide
Snapshot	Exports a PNG of the current view plus all annotations

Only one annotation mode can be active at a time. ESC cancels or exits the current mode.

Annotations are stored per slide name (filename stem). Switching slides hides and shows the correct set; switching back restores them.

Coordinates use OpenSeadragon viewport space (0–1); the SVG overlay is redrawn on every zoom and pan event.

This is independent of TCA overlays served from `tca_url`.

6. Configuration Reference

Variable	Description	Default
EXTERNAL_API_BASE_URL	Pathology API base URL	—
EXTERNAL_API_EMAIL	API login email	—
EXTERNAL_API_PASSWORD	API login password	—
WSI_URL_REFRESH_MINUTES	Re-fetch signed URLs interval	10
SESSION_TTL	Idle session expiry (minutes)	30
ROOT_PATH	Proxy path prefix (e.g. <code>/in/wsi-viewer</code>)	empty
PORT	Set by host (e.g. Cloud Run 8080)	—